

Distribution ∴

① Binomial Distribution ∴

$$f_i = {}^n C_i p^i (1-p)^{n-i}$$

where n is a positive integer
and $0 < p < 1$ and n, p are
two parameters of the distribution.

② Poisson Distribution ∴

$$f_i = e^{-\mu} \cdot \frac{\mu^i}{i!}, i = 0, 1, 2, 3, \dots$$

where $\mu (> 0)$ is the only parameter of the distribution.

③ Uniform distribution ∴

$$f(x) = \begin{cases} 0, & \text{when } x < a \\ \frac{1}{b-a}, & \text{when } a \leq x \leq b \\ 0, & \text{when } x > b \end{cases}$$

where both a and b are real numbers
and a, b are the two parameters of the
distribution.



④ Normal distribution :- $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}$ $-\infty < x < \infty$

m and $\sigma (>0)$ being two parameters of the distribution.

$$X = \begin{bmatrix} & & & \\ m & 0 & 1 & - & - \end{bmatrix}$$

observation :-

① $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}} dx$ [$\because$ put, $z = \frac{x-m}{\sqrt{2}\sigma}$]

$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2} \cdot \sqrt{2}\sigma dz$

$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} dz = \frac{1}{\sqrt{\pi}} \times \sqrt{\pi} = 1$

$\therefore \int f(x) dx = 1$

Mean, Variance and S.D

① Binomial distribution

$$\text{Mean} = m = E(x)$$

$$= \sum_{-\infty}^{\infty} x_i f_i$$

$$= \sum_{i=1}^n i \cdot {}^n C_i p^i (1-p)^{n-i}$$

$$E(x) = x \cdot f(x)$$

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$$= 1 \cdot {}^n C_1 p^1 (1-p)^{n-1} + 2 \cdot {}^n C_2 p^2 (1-p)^{n-2} + 3 \cdot {}^n C_3 p^3 (1-p)^{n-3} + \dots + n \cdot {}^n C_n p^n (1-p)^{n-n}$$

$$= np \left[(1-p)^{n-1} + 2 \cdot \frac{n(n-1)}{2} p^2 (1-p)^{n-2} + 3 \cdot \frac{n(n-1)(n-2)}{3 \cdot 2} p^3 (1-p)^{n-3} + \dots + np^n \right]$$

$$= np [p + 1 - p]^{n-1} = np$$

Binomial
expansion

$(a+b)^n =$

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$$\begin{aligned}
 \text{Again, } E(x^2) &= \sum_{i=1}^n i^2 \cdot nC_i p^i (1-p)^{n-i} \\
 &= 1^2 \cdot nC_1 \cdot p \cdot (1-p)^{n-1} + 2^2 \cdot nC_2 \cdot p^2 \cdot (1-p)^{n-2} + \dots + n^2 p^n \\
 &= np \left[(1-p)^{n-1} + 2(n-1) \cdot p (1-p)^{n-2} + \frac{3(n-1) \cdot (n-2)}{2!} p^2 (1-p)^{n-3} + \dots + n p^{n-1} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= np \left[\left\{ (1-p)^{n-1} + {}^{n-1}C_1 p (1-p)^{n-2} + {}^{n-1}C_2 p^2 (1-p)^{n-3} + \dots + p^{n-1} \right\} + (n-1)p \left\{ (1-p)^{n-2} + \right. \right. \\
 &\quad \left. \left. {}^{n-2}C_1 p (1-p)^{n-3} + {}^{n-2}C_2 p^2 (1-p)^{n-4} + \dots + p^{n-2} \right\} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= np \left[(1-p+p) + (n-1)p \right] \\
 &= np + n^2 p^2 - np^2
 \end{aligned}$$

Therefore,

$$\begin{aligned} V(X) &= \sigma^2 = E(X^2) - \{E(X)\}^2 \\ &= np + n^2 p^2 - np^2 - n^2 p^2 \\ &= np - np^2 \\ &= np(1-p) \\ &= npq \quad \text{where } q = 1-p \end{aligned}$$

$$\left. \begin{array}{l} \text{Thus, mean} = np \\ \checkmark \text{ Variance} = npq \\ \text{S.D} = \sqrt{npq} \end{array} \right\}$$

Poisson Distribution

$$\text{Mean} = m = E(x) = \sum_{-\infty}^{\infty} x_i f_i$$

$$= \sum_{i=0}^{\infty} \frac{e^{-\mu} \cdot \mu^i}{i!}$$

$$= e^{-\mu} \left[1 \cdot \mu + 2 \cdot \frac{\mu^2}{2!} + 3 \cdot \frac{\mu^3}{3!} + \dots \right]$$

$$= e^{-\mu} \cdot \mu \left[1 + \mu + \frac{\mu^2}{2!} + \dots \right]$$

$$= e^{-\mu} \cdot \mu \cdot e^{\mu} = \textcircled{\mu} \left[\because 1 + \mu + \frac{\mu^2}{2!} + \dots = e^{\mu} \right].$$

Again,

$$\begin{aligned} E(X^2) &= \sum_{i=0}^{\infty} i^2 \cdot e^{-\mu} \cdot \frac{\mu^i}{i!} \\ &= e^{-\mu} \left[1^2 \cdot \mu + 2^2 \cdot \frac{\mu^2}{2!} + 3^2 \cdot \frac{\mu^3}{3!} + 4^2 \cdot \frac{\mu^4}{4!} + \dots \right] \\ &= e^{-\mu} \cdot \mu \left[1 + 2\mu + \frac{3\mu^2}{2!} + \frac{4\mu^3}{3!} + \dots \right] \\ &= e^{-\mu} \cdot \mu \left[\left(1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots \right) + \mu \left(1 + \mu + \frac{\mu^2}{2!} + \dots \right) \right] \\ &= e^{-\mu} \cdot \mu [e^{\mu} + \mu \cdot e^{\mu}] = \mu + \mu^2 \quad \checkmark \end{aligned}$$

$$\begin{aligned}
 \text{Var}(x) &= E(x^2) - \{E(x)\}^2 \\
 &= \mu + \mu^2 - \mu^2 \\
 &= \mu
 \end{aligned}$$

Thus, Mean = μ
 Variance = μ .
 S.D = $\sqrt{\mu}$

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